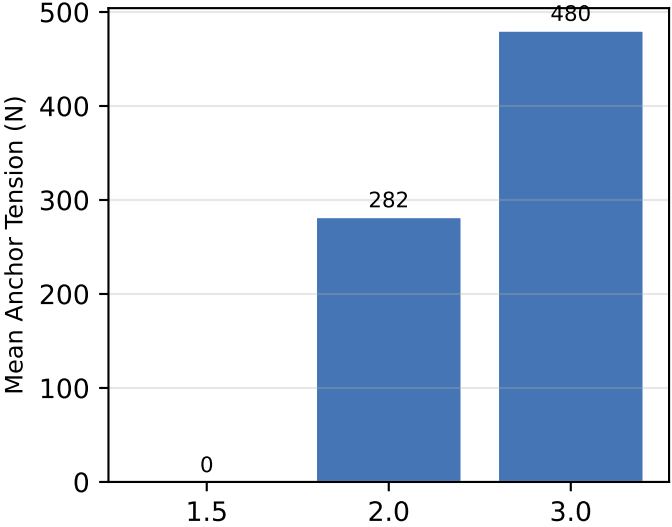
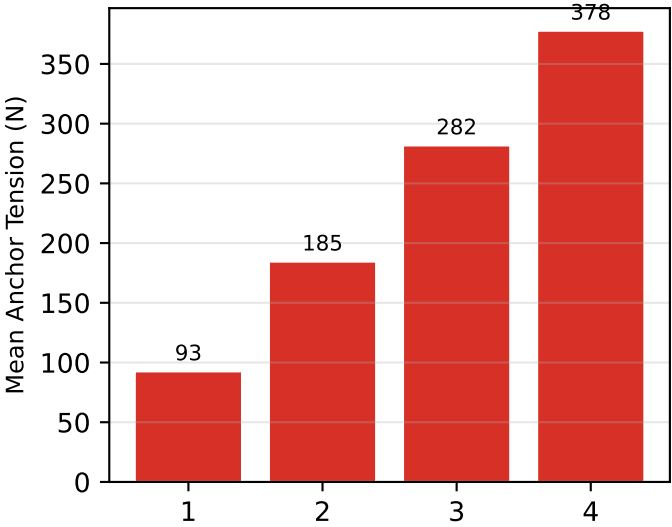


Parameter Sensitivity: Which Design Knob Matters Most?
BEM v2.1 — One-at-a-Time About Baseline Configuration

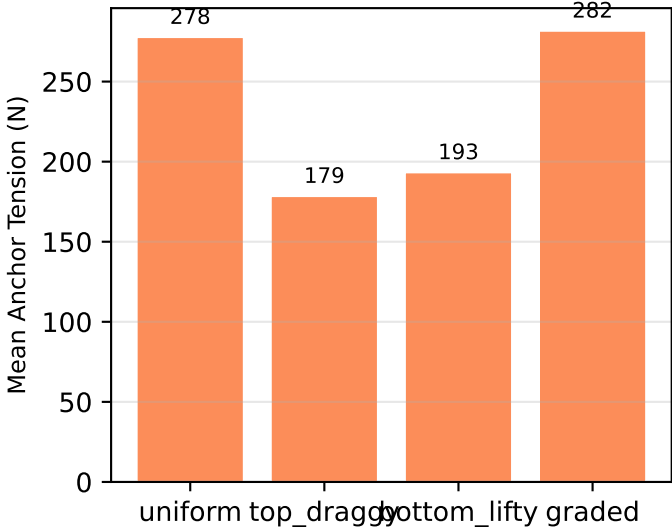
Rotor Radius (m)



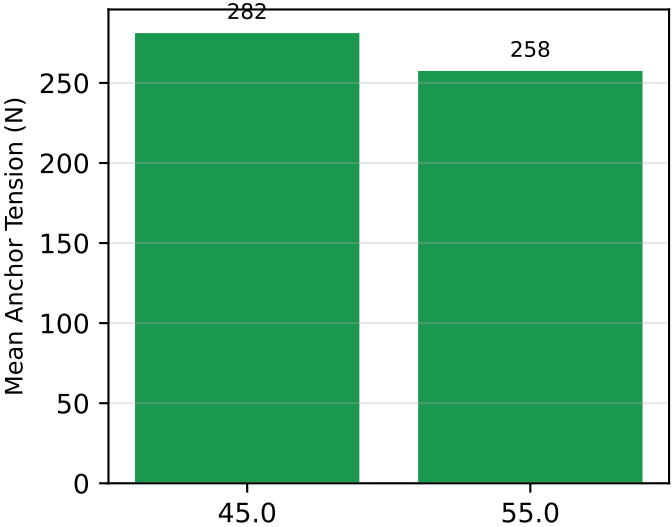
Stack Count



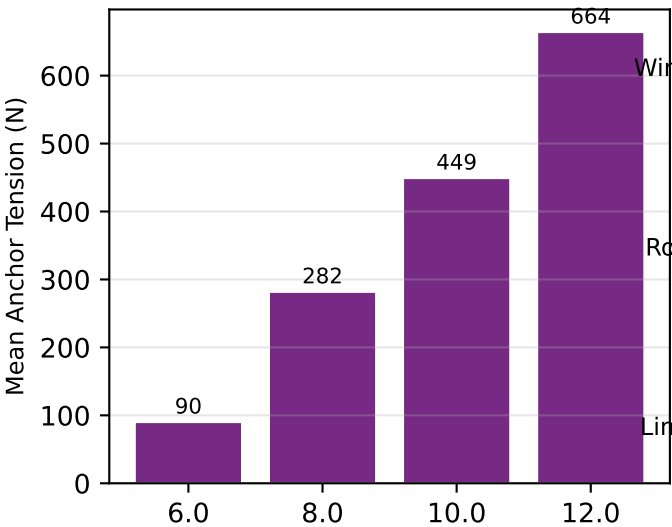
Tilt Profile



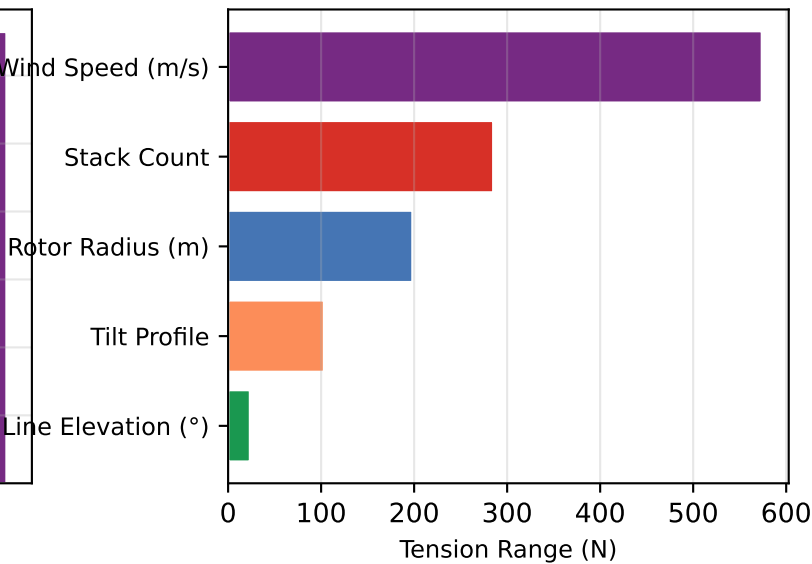
Line Elevation (°)



Wind Speed (m/s)



Relative Sensitivity
(range of mean tension)



IMPLICATIONS: Wind speed dominates ($tension \propto v^2$). Rotor radius is the strongest design lever — doubling from $R=1.5m$ to $R=3.0m$ increases tension by $\sim 10\times$. Stack count has diminishing returns: $N=1\rightarrow 2$ adds more tension than $N=3\rightarrow 4$. Tilt profile and elevation are secondary effects ($<15\%$ variation). Design priority: maximise radius within structural limits, then stack count. Data: BEM v2.1, graded profile baseline, 8 m/s.